

A Cleaner Balance Sheet, or Lasting Capacity? Zambia's Copper Test

Boldwin Mweemba

Independent Analyst, Data Analytics and Economic Research, Lusaka, Zambia

boldwin@aims.edu.gh • September 2026

This piece consolidates a set of independent analytical projects on Zambia's economy available at github.com/BoldwinMax.

Summary

- Copper has broken \$13,000 per tonne. The Kwacha has appreciated 32% since December 2024. Mineral rents have climbed from 3% to nearly 28% of GDP. The recovery is real and the data confirms it.
- The 2020 Eurobond default and its June 2024 restructuring collapsed debt service to exports from a 12.5% peak to under 3%. That improvement is structural, not just cyclical.
- Energy is the constraint copper prices cannot relax: the mining sector consumed 10.6% less electricity in 2024 alone, during a hydrological crisis that pushed Lake Kariba to 10% capacity.
- Revenue is concentrated dangerously: large taxpayers carry 79% of the domestic tax base; one mine (Kansanshi) contributed roughly a third of the entire mining sector's government payments in 2023.
- The window to convert this boom into lasting capacity is now, while the tailwind is still blowing. Energy, tax diversification, and export logistics are the three policy levers that will decide the outcome.

Introduction

Zambia's economy has been on a run. Copper broke \$13,000 a tonne. The Kwacha appreciated 32% between December 2024 and May 2026. Mineral rents climbed from 3% of GDP in 2020 to nearly 28% by 2024. Foreign direct investment surged past 9% of GDP once the Eurobond restructuring closed. Debt service to exports, which peaked at 12.5% in 2019, has fallen under 3%.

By almost every headline number, the recovery is real. The question this piece asks is not whether copper is delivering; the data confirms it is. The question is whether Zambia is converting that tailwind into lasting capacity, or simply building a cleaner balance sheet ahead of the next stress test.

Three things will decide which one it turns out to be: energy reliability, tax predictability, and export logistics. Copper gives Zambia the opportunity. Policy determines whether the opportunity becomes transformation.

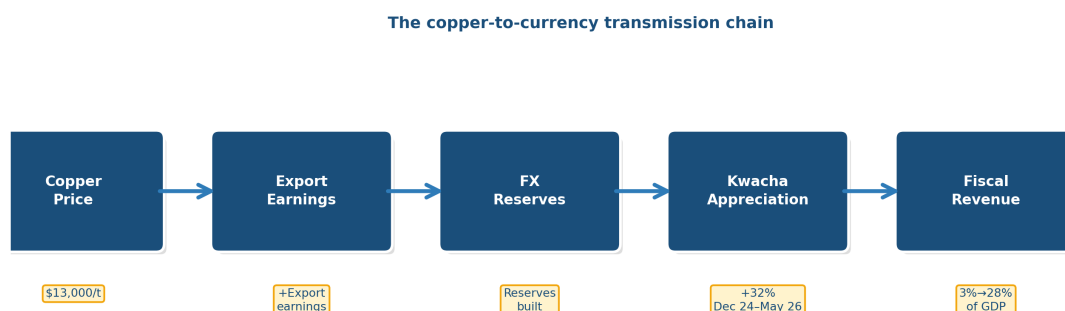


Figure 1. The copper to currency transmission chain. Export earnings growth drives FX reserve accumulation, which in turn supports Kwacha appreciation and expands fiscal revenue. Key figures at each stage: \$13,000/tonne copper price; Kwacha appreciation of 32% (December 2024 to May 2026); mineral rents rising from 3% to 28% of GDP.

What the Data Confirms

The transmission chain from copper to currency is observable and consistent with the mechanism you would expect: as copper prices rose past \$13,000 per tonne, export earnings followed, reserves built, and the Kwacha appreciated in the same window. Foreign exchange reserves alone explain roughly half of the Kwacha's movements historically, which is more than copper prices do directly and is itself a useful reminder that the currency's stability runs through the central bank's reserve position, not commodity prices alone.

The recovery from the 2020 Eurobond default tells a similar two stage story. Copper prices nearly doubled between 2020 and 2022, and mineral rents rose from 3% to 28% of GDP before the formal debt restructuring was finalised. The G20 Common Framework and the June 2024 Eurobond restructuring then locked in the second stage, collapsing debt service to exports from 12.5% to under 3%. Neither stage tells the full story alone: without copper, the debt deal would have landed on harsher terms; without the deal, the copper windfall would have been absorbed by debt service instead of building reserves.

The Constraint the Headline Numbers Don't Show

Copper mining consumes close to 69% of Zambia's national electricity. Reservoir levels at Lake Kariba explain roughly 53% of the variation in load shedding hours, and when the reservoir drops, mine electricity consumption does not hold steady. It fell 10.6% in 2024 alone, during the hydrological crisis.

Zambia has set a production target of 3 million tonnes of copper. Getting there requires a level of consistent power supply the current generation mix (still roughly 85% hydro) has not reliably delivered in drought years. Solar capacity has grown fast, from 88 MW in 2021

to roughly 841 MW now, and that growth has genuinely reduced Zambia's exposure to a poor rainy season. It has not removed that exposure. Solar does not cover the evening peak without storage, and hydro still carries most of the load.

The uncomfortable version of this: a country can have record copper prices and still fail to capture them fully, if the power to run the mines is not there when it is needed.

Energy is the binding constraint copper prices don't relax

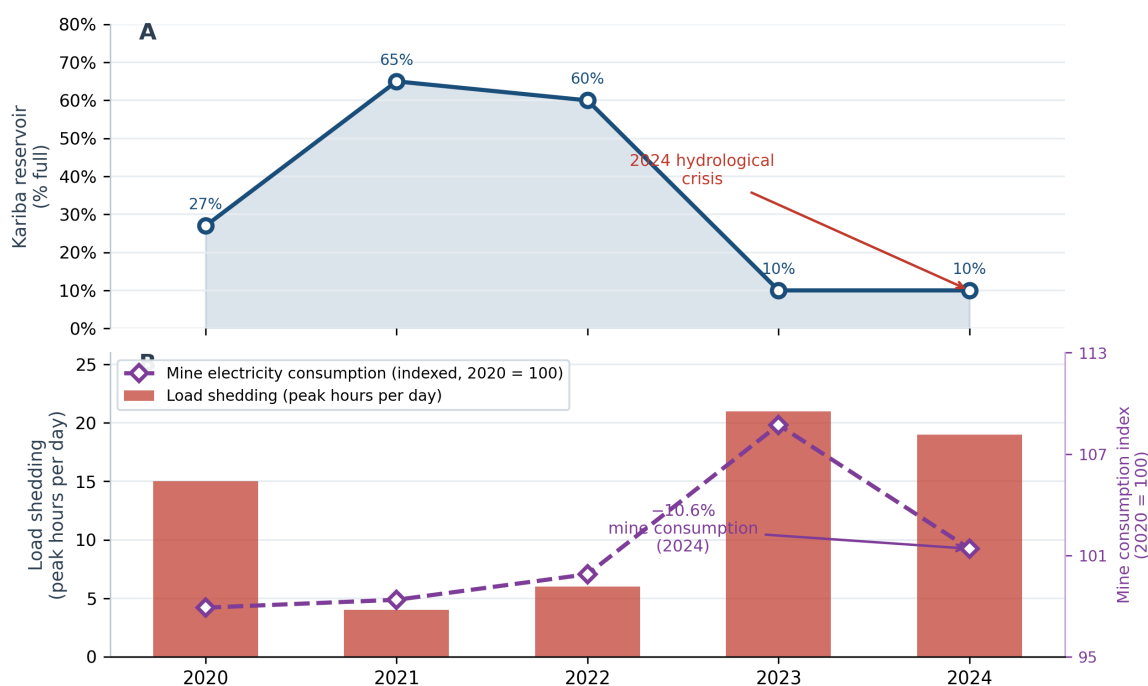


Figure 2. Energy is the binding constraint copper prices do not relax. **Panel A:** Lake Kariba reservoir level (% full), 2020 to 2024, showing the drop to 10% during the 2024 hydrological crisis.

Panel B: Load shedding hours per day (red bars) and mine electricity consumption index (2020 = 100), showing the 10.6% drop in mine power use in 2024 even as copper prices climbed.

The Tax Base Carrying the Weight

The Zambia Revenue Authority is expected to fund up to 80% of the national budget by 2028. Right now, that revenue rests on a narrow base. Large taxpayers alone carry 79% of domestic tax revenue. Private businesses carry 92% of it. Mining is the largest single contributor, and it also received 90% of every tax incentive approved between 2018 and 2023. In 2023 alone, for every kwacha mining paid in tax, it received roughly 8 cents back in incentives. One mine, Kansanshi, paid ZMW 6.3 billion in 2023, which is close to a third of the entire sector's government contribution on its own.

That concentration is efficient to administer. It is not resilient. A revenue base this narrow moves with the fortunes of a handful of firms and a single commodity cycle, which is a fragile foundation for a target as large as funding 80% of the national budget.

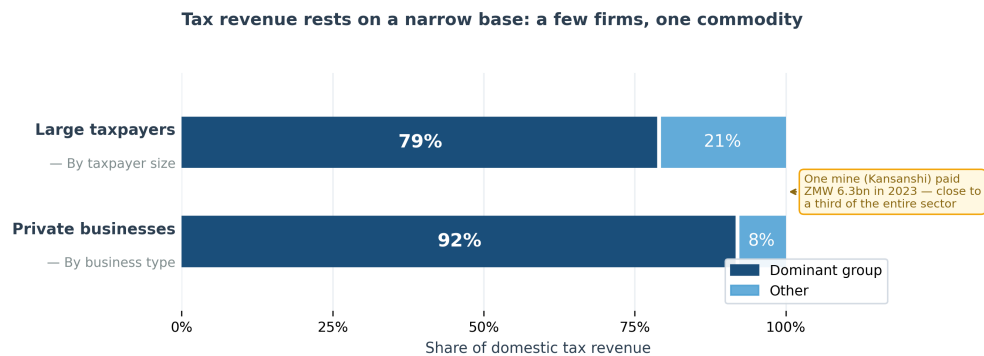


Figure 3.

Tax revenue is concentrated in a handful of firms and one commodity. Large taxpayers carry 79% of the domestic tax base. Private businesses carry 92%. The Kansanshi callout illustrates how thin the diversification is even within the mining sector itself.

Opportunity Is Not the Same as Transformation

None of this is a case for pessimism. Copper is real, and right now it is working in Zambia's favour. The reserves have built. The debt terms have improved. The recovery is measurable, not narrative.

But a boom converts into lasting capacity only if the constraints around it get addressed while the window is open, not after prices turn. Energy reliability determines whether the 3 million tonne production target is achievable or capped regardless of price. Tax base diversification determines whether the ZRA's 80% by 2028 ambition rests on a foundation that can actually hold that weight. Export logistics determine whether the earnings copper generates can actually move to market at the pace the boom demands.

Conclusion

Copper bought Zambia the opportunity. The question is what Zambia does with it, and the answer is almost entirely a function of decisions made in the next two to three years, while the commodity tailwind is still blowing and the fiscal space it has created still exists. The political economy is harder than the economics: tax incentives are easier to grant than to renegotiate, grid infrastructure competes with other budget priorities, and logistics corridors require regional coordination. But the window in which these choices carry the most weight is the one open right now, when reserves are building, debt service is low, and the sovereign has fiscal room to make long term commitments.

The three structural constraints at the centre of this analysis are not abstract concerns. They are the specific bottlenecks that will determine whether the 3 million tonne production target is met, whether the ZRA's 80% budget contribution ambition survives the next commodity softening, and whether the earnings copper generates actually stay in Zambia. The copper price environment is favourable in ways that will not persist indefinitely. Whether the decisions being made right now still look right when the tailwind fades is the question this piece is putting on the table.

References

- [1] London Metal Exchange (LME). *Copper Official Cash Settlement Price*. LME, 2024 to 2026.
- [2] Bank of Zambia (BoZ). *Monthly Exchange Rate Statistics: Zambian Kwacha to US Dollar*. BoZ, Lusaka, 2024 to 2026.
- [3] World Bank. *World Development Indicators: Mineral Rents, FDI, and GDP Series for Zambia*. Washington DC, 2020 to 2024.
- [4] International Monetary Fund (IMF). *Zambia: 2024 Article IV Consultation and Request for Extended Credit Facility*. IMF Country Report, Washington DC, 2024.
- [5] G20 Common Framework for Debt Treatments. *Zambia Debt Treatment Agreement*. Official Communiqué, June 2024.
- [6] Zambia Revenue Authority (ZRA). *Annual Report and Tax Expenditure Statistics*. ZRA, Lusaka, 2021 to 2023.
- [7] Energy Regulation Board (ERB), ZESCO, and Zambezi River Authority. *Electricity Generation, Load Shedding, and Kariba Reservoir Level Data*. Lusaka and Harare, 2020 to 2024.